PROTEIN SYNTHESIS

INTRODUCTION:

The pathway of protein Biosynthesis is called **translation**.

- Translation is the process by which ribosomes convert the information carried by mRNA in the form of genetic code to the synthesis of new protein.
- Translation occurs in cytosol on ribosomes and is guided by mRNA.

The basic plan of protein synthesis in eukaryotes is similar to that in prokaryotes. However, eukaryotic protein synthesis involves more protein components and some steps are more intricate. Some differences are listed in Table 20.3.

Table 20.3: Differences between eukaryotic and prokaryotic protein synthesis

Characters	*Eukaryotes*	*Prokaryotes*
Ribosome	Larger, 80S, consists of 60S and 40S subunits	Smaller, 70S, consists of 50S and 30S subunits
Initiator tRNA	Met-$tRNA_i^{Met}$	f-met-$tRNA_i^{f\ met}$
Start signal	Initiating AUG of mRNA is preceded by a cap, methylguanosyl triphosphate	Initiating AUG of mRNA is preceded by a purine rich sequence
Initiation factors	Contains more intiation factors than do prokaryotes Nine are known and several consists of multiple subunits The prefix eIF denotes a eukaryotic initiation factor	Contain three initiation factors, IF_1, IF_2 and IF_3
Termination factors	Termination is carried out by a single release factor eRF	Termination is carried by three releasing factors RF_1, RF_2 and RF_3

BASIC REQUIREMENTS FOR THE TRANSLATION:

- mRNA
- tRNAs
- Ribosomes
- Energy in the form of ATP and GTP
- Enzymes and specific protein factors, e.g. initiation factors, elongation factors, etc.

STAGES OF EUKARYOTIC TRANSLATION:

There are four major stages in protein synthesis, each requiring a number of components.

1. Activation of amino acids
2. Initiation

3. Elongation
4. Termination.

1. ACTIVATION OF AMINO ACID:

Activation of amino acid takes place in the cytosol. In activation of amino acid each of the 20 amino acids, is covalently attached to their respective tRNA, at the expense of ATP, by **aminoacyl tRNA synthases (AAS)** enzyme (Figure 20.10).

When an amino acid is attached to the tRNA to form aminoacyl tRNA, the latter can recognize a codon on mRNA by virtue of an anticodon in the tRNA structure.

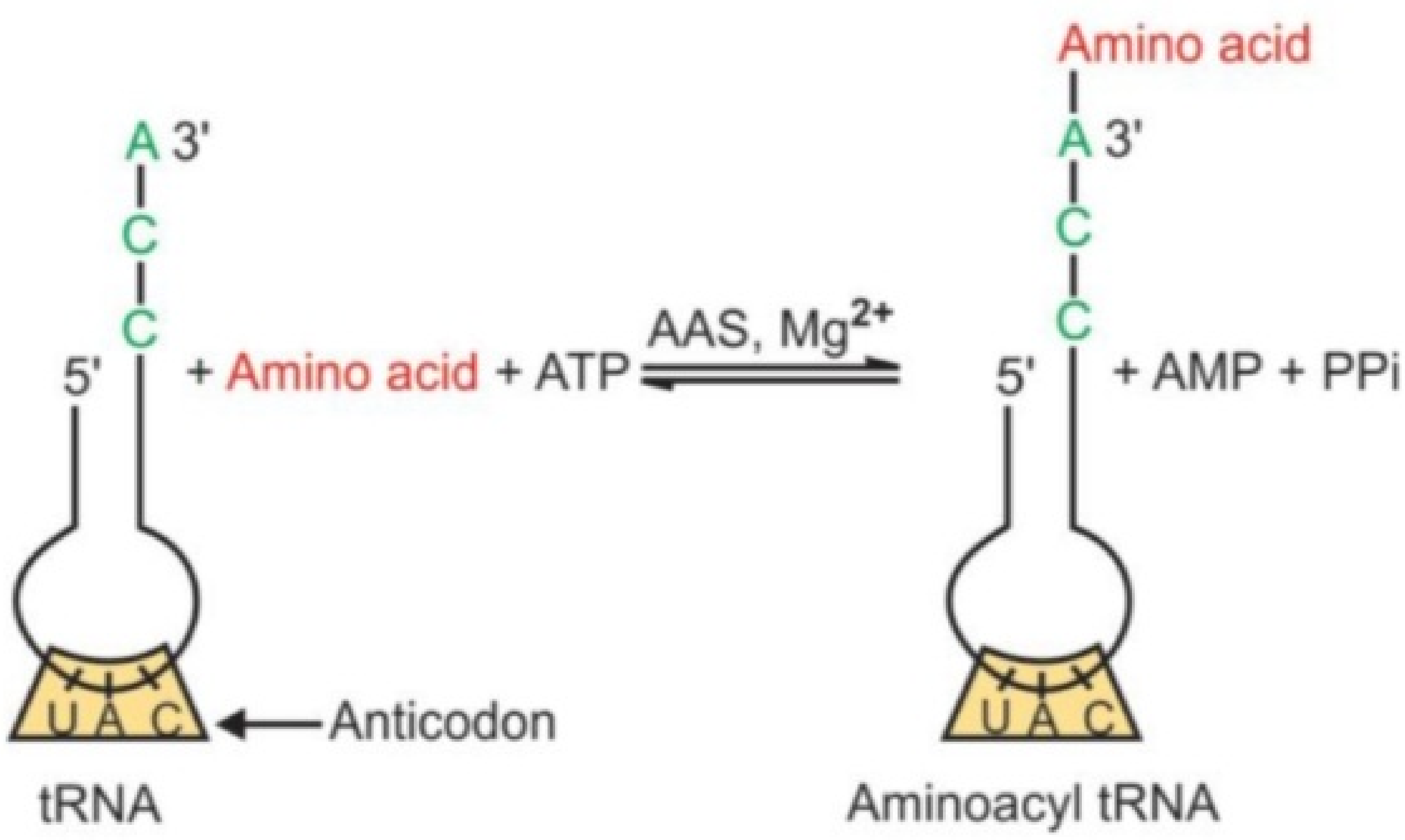

Figure 20.10: Activation of amino acid.
AAS: Aminoacyl tRNA synthase

2. INITIATION (FIGURE 20.11):

Initiation can be divided into four steps:

1) Ribosomal dissociation
2) Formation of 43S pre-initiation complex.
3) Formation of 48S initiation complex
4) Formation of 80S initiation complex.

1) RIBOSOMAL DISSOCIATION:

Eukaryotic ribosomes are 80S particles composed of two subunits of 40S and 60S. The ribosome has two binding sites for tRNA molecules, the **A (aminoacyl)** and **P(peptidyl)** site. 40S subunit of ribosome binds two initiation factors elF3 and elF-1A and dissociates 80S into two subunits 40S and 60S.

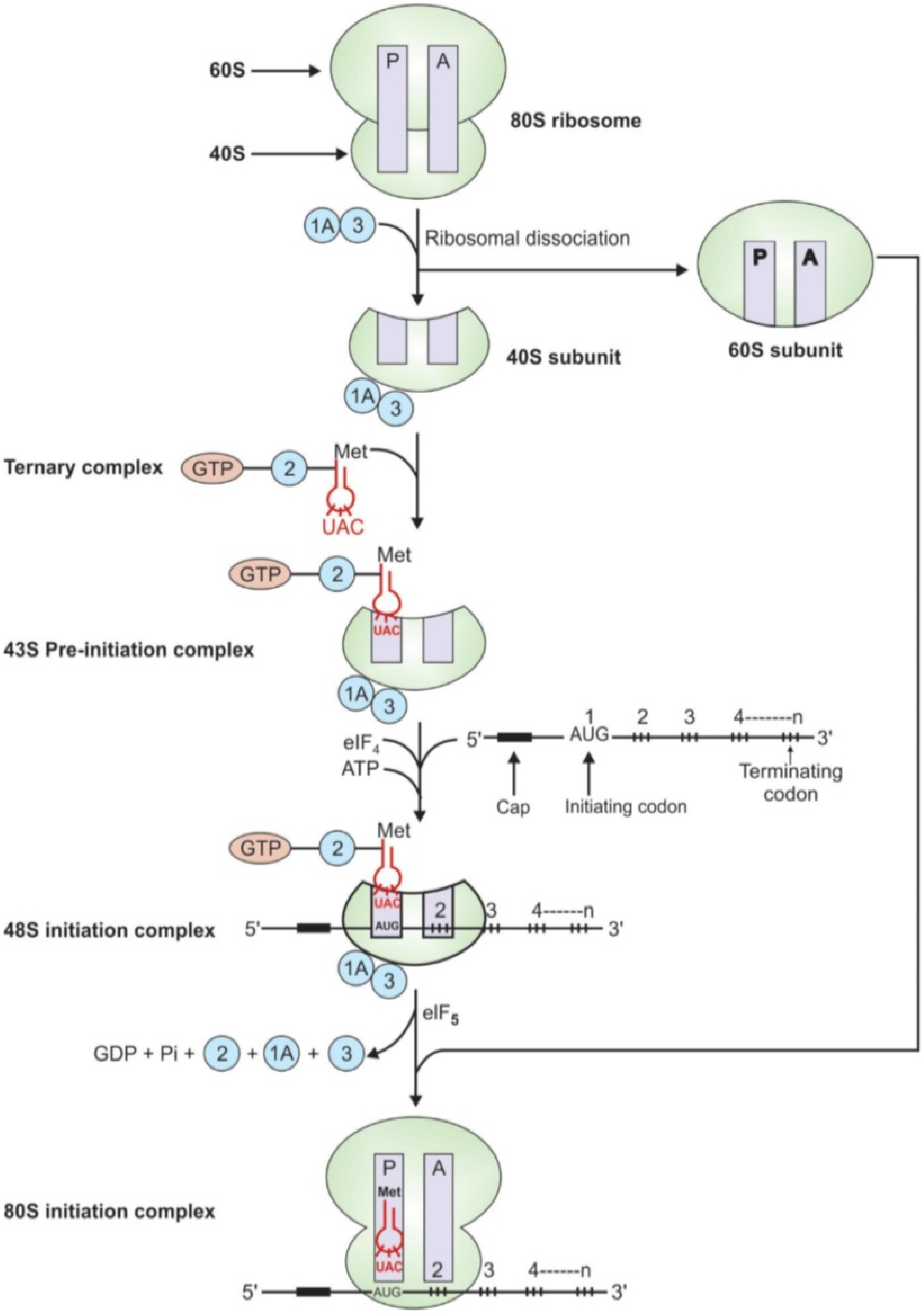

Figure 20.11: Diagrammatic representation of initiation of protein biosynthesis
1A: eIF-IA; 2: eIF_2; 3: eIF_3; CBP: Cap binding protein (eIF_4)

2) FORMATION OF 43S PRE-INITIATION COMPLEX:

- First it involves the binding of GTP with initiation factors elF_2. This complex then binds to Met-$tRNA_i^{Met}$, (a tRNA specifically involved in binding to the initiation codon AUG on mRNA). In prokaryotes, the first amino acid methionine is modified by formylation of its amino group to Nformyl-methionine (f Met) and initiator tRNA is fmet-$tRNA_i^{fMet}$.
- This **ternary (GTP-elF_2-tRNA)** complex binds to the 40S ribosomal subunit to form **43S pre-initiation complex**, which is stabilized by association with elF3 and elF-1A.

3) FORMATION OF 48S INITIATION COMPLEX:

- Binding of mRNA to the 43S pre-initiation complex forms 48S initiation complex.
- The 5' terminals of most mRNA molecules in eukaryotic cells are capped by methylguanosyl triphosphate, which facilitates the binding of mRNA to the 43S pre-initiation complex to form 48S initiation complex.
- The association of mRNA with 43S initiation complex requires **cap binding protein (CBP)** or eIF_4 and **ATP**.

In prokaryotes, a sequence of nucleotide bases on mRNA known as Shine-Dalgarno sequence facilitates the binding of mRNA to the preinitiation complex.

4) FORMATION OF 80S INITIATION COMPLEX:

- Combination of the 48S initiation complex with 60S ribosomal subunit forms 80S initiation complex.
- The binding of the 60S ribosomal subunit to the 48S initiation complex involves the hydrolysis of the GTP bound to elF_2, by elF_5 with the release of the initiation factors bound to the 48S initiation complex. These factors are then recycled.
- At this stage, the Met-$tRNA_i^{Met}$, is on the P site of the ribosome so that its anticodon pairs with the initiating AUG codon on the mRNA and is now ready for the elongation process.

3. ELONGATION (FIGURE 20.12):

Elongation can be divided into three steps

1) Binding of next aminoacyl tRNA specified by the next coding triplet in the mRNA to the A site of ribosome.
2) Formation of peptide bond.
3) Translocation.

1) Binding of aminoacyl tRNA to the A site:

- The next aminoacyl tRNA specified by the next codon in the mRNA is first bound to a complex of elongation factor eEf-1α-containing a molecule of bound GTP
- The resulting aminoacyl tRNA-eEF-1α- GTP complex then allows aminoacyl tRNA to enter the A site on the ribosome with the release of eEF-1α- GDP and phosphate.

2) Formation of peptide bond:

- In the second step of the elongation cycle, a new peptide bond is formed between the amino acids whose tRNAs are located on the A and P sites of the ribosome.
- This step occurs by the transfer of the initiating methionine from its tRNA to the amino group of the new amino acid that has just entered the A site.
- This step is catalyzed by **peptidyl transferase (ribozyme).**
- As a result of this reaction, a dipeptide tRNA is formed on the A site and the "empty" initiating $tRNA_i^{Met}$ remains bound to P site.

3) Translocation:

- In the third step of the elongation cycle the ribosome moves along the mRNA, towards its 3' end, by a distance of one codon (three bases).
- The movement of the ribosome shifts the dipeptidyl tRNA from the A site to the P site, which causes the release of the preceding tRNA, which is empty, from the P site back into the cytosol.
- Now the third codon of the mRNA is on the A site and second codon on the P site. This shift of ribosome along the mRNA is called the **translocation**. It requires elongation factor (translocase) eEF_2.
- The hydrolysis of GTP provides energy for the translocation.

The ribosome with its attached dipeptidyl tRNA and mRNA is now ready for another elongation cycle to attach the third amino acid residue, which proceeds in precisely the same way as the addition of the second. Thus, the process of peptide synthesis occurs until a termination codon is reached.

4. TERMINATION (FIGURE 20.12):

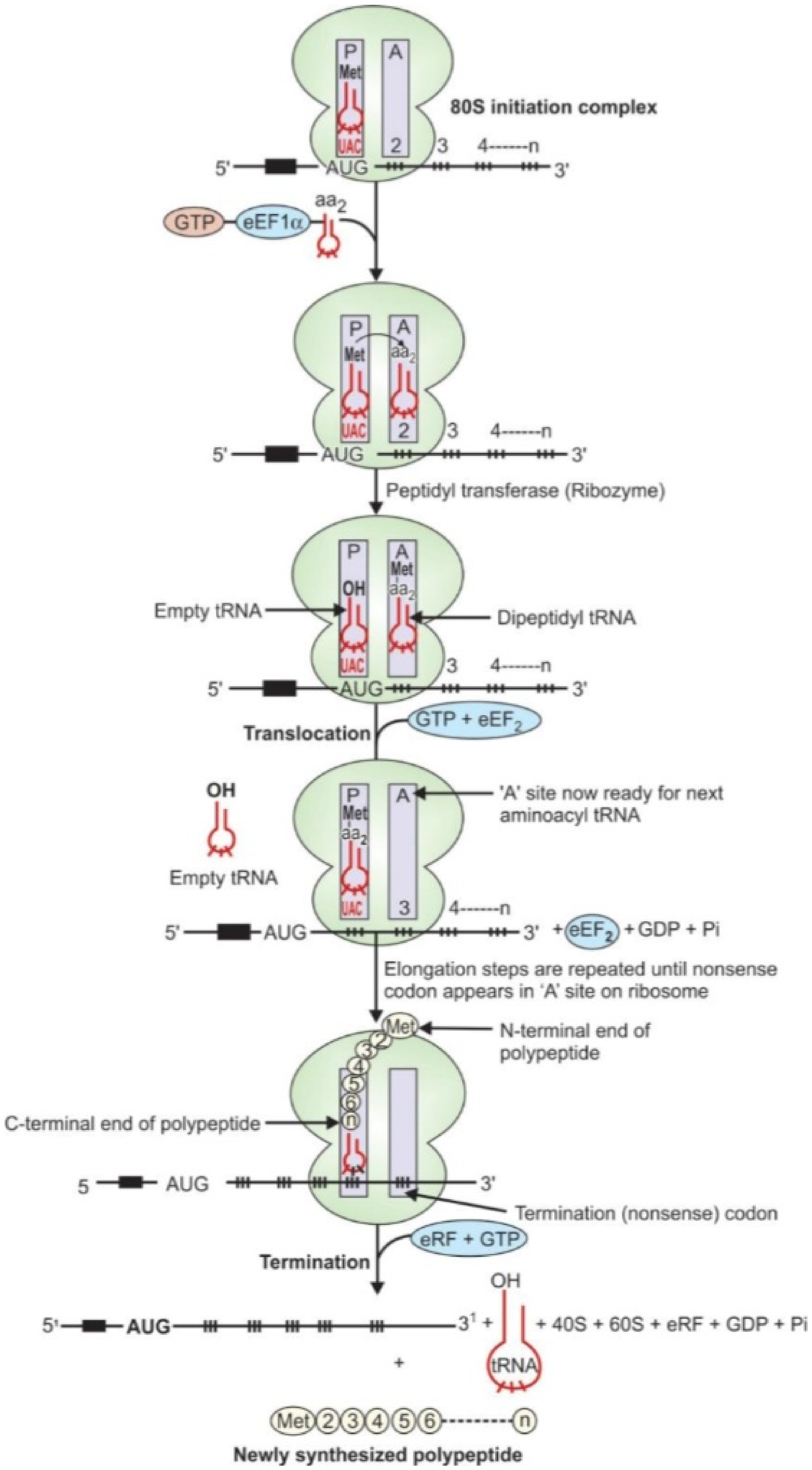

Figure 20.12: Diagrammatic representation of elongation and termination of protein biosynthesis

- The elongation steps are repeated until one of the three (UAA, UAG, UGA) termination or nonsense codon of mRNA appears in the A site.
- Once the ribosome reaches a termination codon, releasing factors are capable of recognizing the termination signal present in the A site.
- Prokaryotes have three release factors, **RF-1**, **RF-2**, and **RF-3** but eukaryotes have only single release factor, eRF.
- Releasing factors in conjunction with GTP and peptidyl transferase hydrolyze the peptidyl tRNA bond (the bond between the peptide and the tRNA), when a nonsense codon occupies the A site.
- This hydrolysis releases the polypeptide and tRNA from P site.
- Upon hydrolysis and release of polypeptide the mRNA is then released from the ribosome.
- Ribosome then dissociates into 60S and 40S subunits, which are then recycled.